Build the complete content operations system

COPY-PASTE BUILD PROMPT

Build Volume 3 as the trusted content layer of the exam platform. A question PDF and a separate solutions PDF must become individually navigable, reviewed questions with faithful mathematics, verified answers and traceable source locations.

Scope and first working slice

Implement an administrator workspace with Uploads, Extraction Review, Item Bank, Paper Builder, Solutions, Tags, Approvals and Corrections. Start with one privately uploaded mixed PDF, one solutions PDF and one custom paper for Tejas. Complete the entire path from upload to a scored practice attempt before adding bulk automation. Use the visual system in Volume 1 and exam configuration contracts in Volume 2.

Build two complementary representations: a faithful source crop for visual inspection and a structured question representation for accessible display, search and analysis. Never claim that an OCR confidence score proves mathematical correctness. Keep uncertain imports in review. Preserve the original source separately from the safe images displayed to students.

Required implementation deliverables

- Create schemas, migrations, authenticated APIs, private object storage, an asynchronous ingestion worker and an administrator review interface. Seed intentionally damaged fixtures as well as a clean sample.
- Create a versioned extraction report showing detected questions, unmatched solutions, suspected numbering errors, missing assets, unresolved answer keys and publication blockers.
- Provide a reproducible import script, a paper validation command and role-based demonstrations. Separate suggested AI output from human-approved content in both database fields and interface labels.
- Return working routes, migration instructions, test evidence and a list of incomplete integrations. A successful animation or mock progress bar is not evidence of a processed PDF.

Non-negotiable behavior

Treat each published paper as an immutable manifest of item versions, solution versions, scoring configuration and release policy. Later edits create new versions. Existing attempts retain their original content and grading lineage. Content publication and solution release are separate server-side permissions.

Use requirements in this volume as executable acceptance criteria. Ask for the actual question PDFs when needed for ingestion, but build and validate the pipeline with synthetic fixtures immediately. Do not fabricate Tejas paper content, official question counts or answer keys to fill gaps.

ACCEPTANCE / PRECAUTIONS

- [ ] An authorized reviewer can upload, correct, approve and publish one real item without editing database records manually.
- [ ] A student attempt receives question assets but cannot retrieve hidden keys or unreleased solutions.
- [ ] A deliberate wrong exponent remains blocked until a reviewer resolves the discrepancy.

Define the upload manifest and intake contract

INGEST-01 | SOURCE INTAKE

Implement a manifest-first intake flow. Capture why a file exists, how it may be used, which paper it belongs to and what the uploader expects before processing begins.

Intake fields and constraints

Collect tenant_id, content_owner_id, original_filename, source_role, language_hint, paper_title, exam_config_version_id, expected_question_count, question_numbering_hint, rights_status and visibility. source_role must be questions, solutions, answer_key or supporting_material. Allow unknown counts during drafting; require a validated paper count before publishing an exam simulation.

Provide a paired upload screen with separate Questions PDF and Solutions PDF slots. Permit questions-only drafts and later solution attachment. Display file sizes, checksum state and upload completion distinctly from extraction progress. Preserve user-entered metadata when a transfer retries. A file completing transfer does not imply that it passed security inspection.

Concrete request and response

POST /admin/uploads with Idempotency-Key and JSON {"source_role":"questions","paper_title":"Tejas Practice 01","exam_config_version_id":"cfg_custom_1","expected_question_count":75,"language_hint":"en","rights_status":"owner_asserted"}. Return {"upload_id":"upl_01","state":"awaiting_bytes","expires_at":"2026-10-12T00:30:00Z","object_key":"private/tenant_a/upl_01"} plus a short-lived upload URL.

After transfer, POST /admin/uploads/upl_01/complete with {"checksum_sha256":"...","byte_size":123456}. Verify actual stored bytes and ownership on the server before creating an ingestion job. Reject a declared checksum mismatch with checksum_mismatch and a restart option. Never accept an arbitrary object_key supplied by the browser.

Intake errors and recovery

- Show distinct errors for unsupported extension, oversized file, incomplete transfer, encrypted PDF, invalid bytes and unauthorized tenant. Avoid one vague upload_failed message.
- Let an encrypted document stay in a restricted draft state; request an authorized unencrypted replacement. Do not store a password in analytics or job logs.
- Use generated storage identifiers. Keep the original filename as escaped display metadata, never as a filesystem path. A duplicate checksum can suggest reuse without exposing another tenant's file.

ACCEPTANCE / PRECAUTIONS

- [ ] A retry returns the same logical upload when the idempotency key and request payload match.
- [ ] A changed payload with an already-used key returns an explicit conflict.
- [ ] The manifest survives a failed transfer and clearly identifies which file must be replaced.

Run durable, idempotent ingestion jobs

INGEST-02 | ASYNCHRONOUS WORK

Process PDFs outside request handlers. Make every stage restartable, observable and bounded so a large import cannot interrupt thousands of active exam attempts.

State machine and durable checkpoints

Use states queued, validating, quarantined, extracting, segmenting, matching_solutions, awaiting_review, failed and cancelled. Publish is a separate action after awaiting_review. Record stage_started_at, heartbeat_at, processed_pages, total_pages, attempt_count, error_code and worker_version. Checkpoint each page and every completed stage; do not restart a 300-page job from page one after a worker crash.

Assign stable IDs to source pages and candidate items using job scope plus source anchors. Enforce database uniqueness for stage outputs. A worker may execute a task more than once; its result must not create duplicate questions or overwrite human edits. New parser settings create a new extraction run linked to the same source.

Polling, retries and cancellation

GET /admin/ingestion-jobs/job_01 returns {"state":"segmenting","pages_done":12,"pages_total":20,"warnings":3,"review_required":true}. Expose events with sequence numbers if using server-sent events. Reconnect from the last event; retain polling as a fallback. The progress percentage must derive from actual completed work, with stage weighting documented.

Retry transient storage timeouts using bounded exponential backoff and jitter. Do not retry permanently malformed files indefinitely. Cancellation should stop future page work, revoke pending tasks and preserve already-created draft evidence. Display cancellation requested until a worker acknowledges it. Offer Resume as a new run rather than silently reviving a cancelled job.

Queue isolation and operational controls

- Place heavy OCR in a separate queue and compute pool from attempt APIs. Set per-tenant concurrent job limits, total page quotas, worker memory ceilings and stage deadlines.
- Capture a sanitized diagnostic bundle with parser version, stage, page index and trace ID. Exclude original question text, user tokens and private object URLs from general logs.
- Create a dead-letter view with Retry stage, Replace source and Download diagnostic options. A privileged operator can inspect failures without granting publication rights.

ACCEPTANCE / PRECAUTIONS

- [ ] Killing a worker halfway through extraction resumes from a recorded checkpoint without duplicated candidates.
- [ ] Cancelling an OCR job does not affect saving student answers.
- [ ] A permanently broken PDF becomes a visible failed job with a bounded attempt count.

Classify native, scanned and mixed pages

INGEST-03 | EXTRACTION TRIAGE

Choose extraction strategies per page and region. A PDF can contain selectable text, scanned pages, embedded images and unreliable hidden OCR in the same document.

Page inventory before recognition

Build a page inventory containing dimensions, rotation, text block count, approximate text coverage, image coverage and extraction warnings. Detect whether native text is usable rather than assuming any text layer is correct. Empty text, repeated replacement characters or wildly mispositioned glyphs should trigger review or OCR fallback. Treat decorative logos and diagrams differently from full-page scans.

Use native extraction for reliable prose and preserve geometric positions. Use OCR only for pages or regions needing it. Official PyMuPDF documentation notes that extracted PDF text can arrive in an unexpected reading order and that OCR is a separate operation [S2, S3]. These facts justify triage; they do not certify any provider's accuracy.

Preprocessing and comparison

Render a safe page image, preserve the untouched original and record any deskew, rotation or contrast adjustment as derived transformations. Detect orientation before segmentation. Expose Original, Processed image, Native text and OCR text tabs to reviewers. Let reviewers compare alternatives where a faint minus sign or overwritten label changes meaning.

Persist OCR engine, engine version, language pack, settings and region confidence. Mark high-risk tokens such as exponents, decimal points, charge signs, inequality direction, chemical subscripts and diagram labels for visual confirmation. Confidence values are provider-specific signals, not comparable correctness probabilities. Do not turn a confidence of 0.98 into an automatic publication decision.

Fallback and cost controls

- When OCR cannot reliably read a question, retain a reviewed source crop and allow manual transcription. Mark semantic completeness separately from visual fidelity.
- Let reviewers rerun only selected pages or regions with a different strategy. Reprocessing must create candidates without destroying prior corrections.
- Estimate OCR workload from page count and region classification. Show an administrator the planned workload and support cancellation before expensive processing. Avoid fixed promises about seconds per page.

ACCEPTANCE / PRECAUTIONS

- [ ] A mixed fixture applies native extraction to one page and OCR to a scanned page.
- [ ] A bogus hidden text layer produces a warning and a side-by-side review.
- [ ] The accepted source image remains available after a failed OCR rerun.

Segment columns, passages and multi-page questions

INGEST-04 | GEOMETRIC SEGMENTATION

Implement layout-aware grouping with editable regions. Question boundaries must follow the source structure instead of a naive split at every numbered line.

Reading-order model

Represent each page as regions with bounding boxes, type, confidence and explicit reading_order. Detect columns using geometry and whitespace, then let a reviewer correct the sequence. A two-column page commonly needs the complete left column before the right column. Preserve reading order as data rather than treating a parser's text sequence as authoritative.

Allow one item to reference several regions and several pages. A stem can begin at the bottom of one page and continue on the next with its options and diagram. Define continuation_links and require a reviewer to confirm their direction. Display clipped boundaries visibly; never lose a line merely because it overlaps an automatically inferred margin.

Shared stimuli and option structure

Create stimulus entities for a common passage, case description, graph or data table. Link child items through stimulus_version_id. Show shared material with every relevant question in student view and preserve it during custom-paper selection. Selecting one child should include the stimulus once, while retaining independent answers and timing.

Create option regions independently from the stem. Support lettered, numbered and unlabeled source options; assign stable internal option IDs only after ordering is verified. A label such as 1 inside a formula must not become an option marker. If options are arranged horizontally or in a grid, keep their original geometry visible during review.

Segmentation editor

- Provide draw, resize, split, merge and reorder controls with keyboard alternatives. Show neighboring questions and page thumbnails while the reviewer edits a boundary.
- Save region polygons or rectangles in a documented coordinate system; include transformation metadata when a page is rotated or deskewed.
- Flag overlaps, uncovered text between questions, orphan diagrams, duplicate region assignment and passages with no children. Some overlaps may be legitimate; allow a reasoned exception instead of silently hiding the warning.

ACCEPTANCE / PRECAUTIONS

- [ ] A two-column fixture imports in the intended order without mixing stems and options.
- [ ] A question spanning two pages renders all lines and its diagram together.
- [ ] A shared graph appears once in paper export and stays linked to each dependent item.

Resolve numbering, headers and section identity

INGEST-05 | IDENTITY RECONCILIATION

Build an explicit reconciliation table. Source numbering is contextual metadata; it is not a globally unique question identifier or a reliable parsing boundary.

Separate four identities

Store item_id for the canonical bank entity, item_version_id for exact reviewed content, source_question_label for what the PDF prints, and paper_display_number for the student-facing position. Preserve labels such as 12(a), Q.07 and section-specific 1 without coercing all of them into the same integer. Use a stable source_local_key containing source file, section and occurrence.

Detect page numbers, running headers, watermarks, document titles and repeated section names using their geometry and repetition. Keep excluded regions inspectable. Do not delete every repeated phrase: a repeated instruction may affect scoring or several questions. Exclude it from the question body only after classifying it as document-level or section-level instruction.

Reconciliation dashboard

Show expected count, detected count, confirmed count, duplicate printed labels, gaps, jumps and out-of-order candidates. Add a table with source page, section, printed label, proposed display number and reviewer status. Offer sort by source reading order, printed label or confirmed paper order; never silently renumber imported sources to hide gaps.

When a solutions PDF restarts numbering for Physics and Chemistry, match within confirmed section identity. Require an explicit reviewer decision when two items share section and label. If a key references Q31 but the question source contains only Q30 and Q32, produce missing_question_for_key rather than creating a guessed item.

Instruction capture and tests

- Extract source instructions into a separate reviewed record. Compare them with the selected configuration in Volume 2. A mismatch blocks simulation publication until the intended rule source is resolved.
- Allow intentional custom-paper numbering to differ from original labels. Display original labels in the review screen and provenance exports, without confusing students with two large competing numbers.
- Preserve Roman numerals and subsection boundaries. Treat a number in a chemical formula or list of statements as content unless layout and reviewer evidence identify a question boundary.

ACCEPTANCE / PRECAUTIONS

- [ ] A footer numbered 12 does not create a twelfth question.
- [ ] Two section-specific Q1 items remain distinct and receive the correct solutions.
- [ ] An unexplained count mismatch prevents simulation publication with a readable reason.

Preserve visual fidelity with safe question assets

INGEST-06 | CROP AND RENDER CONTRACT

Use source crops as a fidelity safeguard, with sufficient resolution and controlled delivery. A crop must be readable, complete and associated with its exact source anchor.

Asset generation and storage

Generate reviewed question crops from safe page renders. Store original dimensions, crop coordinates, rendering scale, page rotation, image checksum and asset_version_id. Use a high-quality master plus responsive derivatives sized for real display widths. Avoid scaling every crop to a fixed tiny image that makes superscripts unreadable on a phone.

For a multi-region item, compose crops in confirmed reading order with clear spacing and no duplicated line at a page join. Keep the full-page context accessible to reviewers. A cropped question containing an adjacent answer, handwritten solution or key annotation must be rejected or recropped before student delivery.

Student delivery and accessibility

Display crops at their natural aspect ratio with zoom and horizontal panning for wide diagrams. Keep answer controls outside the image so touch selection remains reliable. Offer a structured-text alternative where reviewed transcription exists. Crop-only items need reviewed descriptive text for important visual content; image-based content is not automatically accessible.

Deliver assets through an authorization-aware gateway or time-limited URLs scoped to the permitted attempt. Build a question-only asset namespace separate from solutions and keys. Preload a small navigation window according to Volume 5 performance budgets. Show an asset_failed state and Retry image control while retaining the student's response and timer state.

Fidelity review rules

- Compare each crop with its source at common desktop and mobile widths. Inspect clipped radicals, graph axes, dotted lines, thin minus signs and pale labels.
- When rasterizing vector graphs, choose reviewed resolution and contrast. Do not use image enhancement that invents labels or changes geometry.
- Cache by immutable asset version. A later corrected crop obtains a new key; purge obsolete delivery references when a paper is withdrawn, without mutating the original evidence.

ACCEPTANCE / PRECAUTIONS

- A full radical, numerator and denominator remain legible at the supported mobile width.
- A failed asset fetch gives a recovery control without clearing the chosen answer.
- A student cannot guess a question asset URL and retrieve a solution crop.

Create structured mathematics, chemistry and diagrams

INGEST-07 | SEMANTIC AUTHORING

Build a structured authoring model that can preserve equations and diagrams while enabling accessible reading, search and reliable responsive layouts.

Block model and supported notation

Represent content as ordered blocks: paragraph, inline_math, display_math, figure, table, list and stimulus_reference. Store mathematical source separately from plain prose. Define a restricted rendering profile for reviewed LaTeX or MathML and record renderer_version. MathML is a web standard for mathematical notation [S4]; selecting it does not remove the need to test actual browsers and assistive technology.

Support fractions, roots, matrices, piecewise expressions, vectors, limits, integrals and reaction notation required by the supplied papers. Validate syntax before saving and preview the exact student renderer. Disable dangerous extensions, embedded HTML and arbitrary macros. An equation that renders without an error can still contain a wrong exponent; retain source comparison.

Chemistry and figure handling

Keep chemical charges, isotopes, subscripts, reaction arrows and bond labels visually distinct. Accept reviewed SVG or raster diagrams; sanitize SVG and strip scripts, external references and event handlers before use. Preserve the original image as private evidence. Do not redraw a molecule using AI and assume it retains stereochemistry.

For tables, store rows, columns, merged-cell metadata and meaningful headers. Provide captions and descriptions for graphs without revealing solutions. A diagram description should communicate labels and relationships needed to attempt the question, not announce the intended inference. Review alt text with the same academic care as the stem.

Round-trip validation

- Provide a diff view between source crop, structured rendering and previous item version. Highlight changed mathematical tokens and replaced figure assets.
- Use plain text for search indexing, with mathematical tokens retained where possible. Do not render raw OCR output as HTML.
- Maintain render fallback: if the semantic renderer fails, display the already-approved fidelity asset and log the error. Publish only when the fallback and answer controls have been verified.

ACCEPTANCE / PRECAUTIONS

- A fixture containing a matrix, chemical charge and labeled graph renders without missing symbols.
- Sanitized SVG cannot fetch an external URL or execute embedded script.
- A semantic rendering error falls back to a reviewed crop while preserving attempt state.

Establish the canonical item and provenance schema

INGEST-08 | DATA CONTRACT

Make provenance part of the database contract rather than a footnote. Every visible stem, option, figure and solution must be traceable to reviewed source evidence.

Core entities and required fields

Create SourceFile, ExtractionRun, SourceRegion, Item, ItemVersion, StimulusVersion, SolutionVersion, AnswerKeyVersion, Paper and PaperVersion entities. Item holds durable identity and ownership. ItemVersion holds content_hash, blocks, options, response_type, source_regions, editorial_status, approval_record and created_at. Use UUIDs or equivalent opaque identifiers; never derive authorization from sequential numbering.

Example item version: {"item_id":"itm_42","item_version_id":"iv_3","response_type":"numeric","stem_blocks":[{"type":"paragraph","text":"Find the value."}],"source_regions":[{"source_file_id":"src_1","page_index":4,"box":[0.12,0.20,0.88,0.42],"coordinate_space":"normalized_top_left"}],"editorial_status":"review_required"}. Specify that page_index is zero-based; show human-facing page 5 in the interface.

Provenance and derived content

For each source anchor retain checksum, page dimensions, rotation and transformation ID. For manual transcription, record who transcribed it, who verified it and the verification timestamp. For AI suggestions, retain provider, model identifier, prompt template version and acceptance decision without logging secrets. A provenance record should distinguish a source fact from an editorial improvement.

Link explanations, corrected diagrams and tag assignments to their own versions. Do not attach an anonymous JSON answer field directly to a mutable question row. Keep academic evidence and presentation assets separable so changing a thumbnail does not unnecessarily change the grading key.

Validation and confidentiality

- Validate fields by response type. An ordinary single-choice key has one approved correct option ID. Approved profile-specific anomaly policies may accept multiple valid alternatives, all options or a dropped-item branch while the UI stays single-select. Require a reviewer to verify each exception against its source and frozen configuration.
- Apply tenant ownership checks on every nested reference. A user must not attach another tenant's private source region or solution by supplying its ID.
- Create public question DTOs with an explicit allowlist. Source credentials, private notes, answer keys, approval comments and unreleased solution IDs must not enter attempt payloads.

ACCEPTANCE / PRECAUTIONS

- [ ] A reviewer opens a stem block's source anchor and sees the correct page and region.
- [ ] A cross-tenant nested asset reference returns authorization failure.
- [ ] The public question response has no hidden grading fields, even before client filtering.

Match separate solutions without guessing

INGEST-09 | SOLUTION RECONCILIATION

Build solution matching as a reviewable mapping problem. Question numbering, section identity and source evidence must outweigh a model's confident narrative.

Candidate matching and evidence

Ingest the solutions PDF through the same security and layout pipeline, with source_role=solutions. Extract candidate explanation regions and key references. Generate proposed links using section identity, printed labels, stem phrases and known cross-references. Display the evidence for every proposal. Store match_method, supporting_anchors, reviewer_decision and ambiguity_reason; do not expose a fabricated probability as certainty.

Allow one solution to cover a shared stimulus plus several items, several solution regions to explain one item, and alternative approved methods. Keep the answer key as its own record. A paragraph containing a worked calculation can suggest an answer, but must not automatically establish the grading key.

Reconciliation interface

Show Question, Proposed solution and Source context in three panes, with linked page navigation. Provide Accept link, Choose another, Split solution, Merge regions and Mark missing controls. Show unresolved and multiply matched items in dedicated queues. Keyboard shortcuts may speed review, but a bulk acceptance action must display how many ambiguous mappings it would affect.

Example mapping: {"item_version_id":"iv_3","solution_version_id":"sv_8","source_label":"Q42","section_id":"physics","status":"proposed","evidence":["same_section","same_label"]}. Publication requires status=approved. A reviewer rejecting a mapping must leave a reason such as numbering_restart or neighboring_solution so later reruns do not keep repeating the same suggestion.

Missing and inconsistent solutions

- Allow publication without explanations only when the configured product policy explicitly permits it and the approved key exists. Display Solution not yet available after permitted release.
- If the key says B and the explanation concludes C, open an academic conflict. Require resolution by a designated subject reviewer; do not silently select the longer explanation.
- Handle out-of-order pages, omitted question numbers and duplicate labels without overwriting links. Preserve original solutions and editorial supplements with clear provenance.

ACCEPTANCE / PRECAUTIONS

- [ ] A solutions document with restarted section numbering maps to the correct subject items.
- [ ] An explanation-key conflict blocks the affected scored paper.
- [ ] A missing solution is reported explicitly and never replaced by an unreviewed AI explanation.

Validate typed answer keys against exam configuration

INGEST-10 | GRADING INPUTS

Define precise, typed key records and bind their validation to the versioned scoring configuration in Volume 2. Do not infer exam rules from a source PDF title.

Typed answer contract

Use key variants single_choice, multi_choice, integer, decimal, numeric_range and unscored_review. Store the accepted option IDs rather than letters, because paper assembly can change displayed option order. For numeric variants retain the authored exact value or accepted interval as a decimal string, together with units policy and normalization_profile_id.

Example {"key_version_id":"kv_4","item_version_id":"iv_3","kind":"integer","accepted_values":["24"],"normalization_profile_id":"np_custom_integer_v1","review_status":"approved"}. Reject fractional accepted values under an integer response profile. Preserve negative values and zero where the chosen configuration allows them. Do not add tolerance, rounding or unit stripping because another exam previously used it.

Validation beyond syntax

Check that each approved item has an approved key compatible with the paper's response and scoring profiles. Require a subject reviewer to verify the value against both question and explanation. Record verification evidence and any dispute. Never assume OCR recognition of an answer-key table is reliable enough for direct production grading.

Create boundary fixtures from the exact configured numeric policy: accepted answer, rejected nearby value, leading or trailing zeros, negative sign, locale separator, blank answer and forbidden characters. The server and client must use the same documented normalization behavior, but the server remains authoritative. Show raw submitted response beside normalized response during support investigations.

Paper-level grading integrity

- Validate item points, section membership and key completeness before publication. If a custom question has no key, mark it unscored and exclude it transparently from the available-score denominator.
- Run a dry grading pass with known correct, wrong and blank answers for every response type in the paper. Compare totals with a reviewer-approved expected result.
- Retain answer_key_version_id in every grading run. A corrected key creates a new version and enters the correction workflow on page 20; it does not rewrite historical results invisibly.

ACCEPTANCE / PRECAUTIONS

- [ ] An integer paper rejects an incompatible decimal key at validation time.
- [ ] Option reordering preserves grading correctness through stable option IDs.
- [ ] A dry-run report proves totals and denominator behavior for correct, wrong, blank and unscored items.

Separate the item bank from immutable paper versions

INGEST-11 | CONTENT VERSIONING

Allow reusable academic content while freezing everything a student actually attempted. Canonical bank edits and published paper changes must follow different rules.

Bank entity versus delivery snapshot

The bank Item points to its latest editorial version and may accumulate tags, alternate solutions and corrected assets. A PaperVersion references exact item_version_ids, exact key versions, exact stimulus versions and a frozen scoring_config_version_id. Add display order, option order, section boundaries, release policy and a content manifest hash. Never render an old attempt using the current Item row.

Example paper manifest {"paper_version_id":"pv_7","exam_config_version_id":"cfg_9","items":[{"item_version_id":"iv_3","key_version_id":"kv_4","display_number":1,"section_id":"math","option_order":[]}],"publication_state":"scheduled","manifest_hash":"..."}. Validate the full manifest and sign or hash it before publishing. Keep version history visible to administrators.

Safe updates and reuse

Creating a new item version should show impacted draft papers and published papers. Drafts may explicitly adopt the update after validation. Published papers need a replacement version or a documented correction action. Editing a title or tag must not silently replace the stem that an active student sees. Distinguish display-only metadata from grading-affecting changes.

When reusing an item in several papers, preserve independent paper context: allocated marks, response policy, display number and shared-stimulus placement. Verify compatibility with each exam configuration. Reuse must not imply that every exam supports the same response type or partial-credit rules.

Concurrency and lifecycle

- Use optimistic locking with version or ETag on editorial updates. Return conflict with a side-by-side diff when two reviewers edit the same item; do not use silent last-write-wins.
- Support draft, approved, scheduled, published, withdrawn and archived paper states. Withdrawal prevents new attempts while defining what happens to already-started attempts through the operations policy.
- Keep content hashes on item and manifest versions. Use them for integrity checks and cache identity; do not use identical hashes as a reason to merge unrelated ownership records automatically.

ACCEPTANCE / PRECAUTIONS

- [ ] A completed attempt displays its original stem after the bank item is corrected.
- [ ] A competing edit receives a conflict instead of overwriting the first reviewer.
- [ ] A manifest containing an incompatible response type cannot reach published state.

Build a verified, versioned tagging ontology

INGEST-12 | QUESTION-WISE TAGS

Design tags for reliable learning analysis and search. Tag suggestions are useful, but official-looking labels and difficulty values must have evidence and editorial status.

Taxonomy and assignment contract

Create hierarchical Subject, Chapter, Topic and Subtopic entities with stable IDs, aliases, syllabus_reference and taxonomy_version. Add non-hierarchical skills such as interpretation, calculation, conceptual reasoning, graph reading and multi-step synthesis. Permit multiple skill tags while keeping a clear primary chapter for supported analysis. Include prerequisite tags and source-specific labels without mixing them into the main taxonomy.

Store tag assignments separately: {"item_version_id":"iv_3","tag_id":"topic_limits","status":"human_verified","assigned_by":"reviewer_2","taxonomy_version":"tax_1","evidence_note":"Uses a left-hand limit"}. Maintain statuses suggested, verified, rejected and deprecated. Reject a tag without losing the model suggestion history. Students should see only approved public tags.

Difficulty and cognitive demand

Keep editorial difficulty distinct from empirical difficulty. A subject reviewer can estimate easy, medium or hard with a brief rationale; the analytics system later computes response statistics for a specified cohort and exposure period. Do not call a question nationally top 1 percent difficult from a small local sample. Store the method and cohort beside every statistic.

Tag common traps only when supported by the question: sign error, unit conversion, domain restriction, extraneous root or option misread. Do not label an individual student as careless from a single wrong response. Connect these tags to cautious explanations in Volume 4, where limitations and minimum sample thresholds govern analytics.

Ontology operations

- Provide merge, rename, alias and deprecate operations with impact previews. A rename should not invalidate attempts; a chapter restructuring needs a new taxonomy version and explicit analytics migration.
- Allow reviewed bulk assignments from CSV or JSON with dry-run validation. Show unresolved aliases, cross-subject conflicts and missing prerequisites before committing.
- Offer a taxonomy coverage dashboard showing untagged items, single-review tags and conflicting assignments. Make academic review queues filterable by subject and reviewer expertise.

ACCEPTANCE / PRECAUTIONS

- [ ] An AI-proposed chapter remains hidden until a reviewer approves it.
- [ ] Renaming a chapter keeps its stable identity and historical reports readable.
- [ ] Empirical difficulty displays its cohort and does not replace the editorial rating silently.

Design a fast, precise academic review workspace

INGEST-13 | REVIEW INTERFACE

Build a review screen that lets administrators correct difficult content accurately without returning to database tools or reconstructing evidence from memory.

Layout and interaction

Use a clear blue-themed workspace with a left candidate list, central student preview and right source evidence panel. Keep question status, current paper section and unresolved blocker count visible. Provide expandable tabs for source crop, structured text, options, key, explanation, tags and history. Remember panel sizes per reviewer without hiding critical warnings.

Include keyboard navigation, undo within the current editing session, explicit Save draft and a visible unsaved-changes indicator. An autosave may persist a draft but must not imply approval. When a browser refresh occurs, recover the last server-saved draft and tell the reviewer whether any local edit was lost.

Focused review tools

Highlight tokens most likely to change meaning: minus signs, exponents, roots, inequality symbols, decimal points and diagram labels. Let reviewers mark each discrepancy resolved with a chosen source region. Display option ordering and key selection together, because a swapped option creates a grading error even when the stem is correct.

Offer next unresolved item, previous item, open neighboring source page and compare previous version. Include a compact paper strip for count and numbering reconciliation. A reviewer should never need to approve a question while its solution or diagram is offscreen without an accessible evidence link.

Quality gates and efficient collaboration

- Show blockers by category: source fidelity, response type, answer key, solution mapping, tag completeness, rights and release policy. Clicking a blocker focuses the exact field and explains the required resolution.
- Allow academic comments and assigned review tasks. Keep internal notes out of student payloads. Track resolved comments with the version they referred to so a later edit does not falsely inherit their resolution.
- Provide bulk operations only for safe metadata changes. Bulk approval of academic content requires explicit selection, an impact summary and recorded reviewer responsibility. Do not add a one-click approve-all button to bypass missing keys or conflicts.

ACCEPTANCE / PRECAUTIONS

- [ ] A reviewer fixes a clipped graph, updates its key and verifies the resulting student preview in one workspace.
- [ ] A stale review comment stays associated with the version it reviewed.
- [ ] A save-draft action never changes a candidate to approved or published.

Implement editorial roles and approval accountability

INGEST-14 | GOVERNED WORKFLOW

Define capabilities by action and content scope. The person uploading a document should not automatically gain authority to publish it or release its solutions.

Roles and action permissions

Implement uploader, content_editor, subject_reviewer, publisher and operations_admin capabilities, with tenant and subject scope. An uploader submits sources and views their jobs. An editor repairs extraction and proposes tags. A subject reviewer approves academic content and keys. A publisher validates paper completeness and schedules release. An operations administrator manages queues and access without automatically passing academic review.

For a small initial team, permit one user to hold several capabilities while preserving separate actions and audit records. The production policy should support independent review of scoring-affecting changes. Avoid hardcoding a single global admin=true flag that gives every support user access to all private papers and answer keys.

Approval evidence

Create approval records containing target_version_id, review_scope, decision, actor_id, timestamp, unresolved_exceptions and evidence_notes. Scope can be stem_fidelity, key_correctness, solution_correctness or publication_readiness. A paper is ready only when the configured required scopes are complete on the exact referenced versions.

If an editor changes an approved key or stem, invalidate the affected approval scopes and return the new version to review. A harmless internal note should not reset academic approval. Encode which fields trigger which invalidations so the interface can explain the consequence before saving.

Audit and delegated work

- Record upload completion, extraction run creation, candidate edits, mapping decisions, approvals, publications, withdrawals and key corrections in an append-only audit log.
- Provide reviewer assignment, workload queues and due dates. Reassignment must not delete the first reviewer's work or grant the new reviewer broader tenant permissions.
- Show publication errors by missing capability or unresolved academic gate. Enforce checks on the server, including scheduled background publication. Hiding a button in the interface is not sufficient authorization.

ACCEPTANCE / PRECAUTIONS

- [ ] An uploader calling the publish endpoint receives a permission error.
- [ ] Editing an approved correct answer invalidates key approval on the new version.
- [ ] A scheduled publication job honors the same readiness checks as manual publication.

Create Tejas custom practice without mislabeling simulations

INGEST-15 | CUSTOM PAPER BUILDER

Build a custom-paper mode for chapter-limited practice and unusual response requirements. Keep its rules explicit so learners can distinguish personal practice from an exam-format simulation.

Custom paper configuration

Allow Tejas or an authorized teacher to select verified chapters, item count, response types, editorial difficulty mix, available time and solution-release timing. Provide a reusable preset for a chapters-only integer practice paper when suitable verified items exist. Do not hardcode Tejas's earlier syllabus as a permanent fact; collect the current chapter selection in the builder.

Assign mode=custom_practice and show a clear label throughout the attempt and report. If the user selects only integer items or changes the total time, do not call the resulting paper an exact JEE Main simulation. The simulated exam modes in Volume 2 must pass their own complete configuration validation.

Assembly algorithm and constraints

Filter only approved, authorized and renderable item versions. Exclude withdrawn items, incompatible response types, unresolved keys and duplicated source occurrences. Handle shared passages as grouped constraints. Let the user choose whether previously attempted items may appear; explain that repeated exposure changes the meaning of percentile comparisons.

Use a deterministic seed for reproducibility, store the selection algorithm version and show a preflight summary with subject counts, chapter coverage, estimated time and marks. If 75 qualifying questions are requested but only 48 exist, present the shortage and available choices. Do not generate or relabel 27 unreviewed questions silently.

Teacher and learner safeguards

- Allow a teacher to preview, save a draft and assign the frozen paper version. Keep answer keys unavailable to the learner until the selected release condition.
- Show where practice rules differ from the chosen exam configuration, including response format and duration. Preserve scoring policy in the manifest rather than reading the latest preset at attempt time.
- Prevent near-duplicate stems from flooding a paper where practical. Use similarity suggestions for review, then retain intentional variants when they test distinct learning goals.

ACCEPTANCE / PRECAUTIONS

- A chapter-limited integer paper is labeled custom practice in its result.
- Insufficient inventory gives an honest shortage report instead of unreviewed filler.
- The same seed and manifest inputs reproduce the same ordered item selection.

Support multilingual content and accessibility review

INGEST-16 | LANGUAGE VARIANTS

Treat translated content as reviewed variants linked to the same academic item. Translation must preserve mathematics, option meaning and grading alignment.

Language variant model

Create localized item versions with locale, source_language, translation_method, reviewer_id and equivalence_status. Store language-independent option identities while allowing reviewed localized labels and text. A language switch must preserve the selected internal option ID. Link each variant to exact figure and stimulus versions; shared graphics can remain language-neutral only when their labels are understandable.

For bilingual source PDFs, identify whether the source presents paired sentences, separate columns or repeated full questions. Do not import the two language presentations as two distinct academic questions. Provide a bilingual evidence view that reveals differences and missing clauses before approving equivalence.

Translation and screen-reader behavior

Use AI translation only as a draft suggestion. Require a competent language and subject reviewer for graded material. Preserve mathematical symbols, units, proper nouns and technical terminology; mark disputed terminology explicitly. Do not promise every language for every uploaded document. Display a missing_variant state and provide the available reviewed language choices.

Supply headings, meaningful table headers, keyboard-accessible image zoom and reviewed descriptions for essential diagrams. Test source-order reading and focus movement in the attempt preview. Language changes should update document language metadata so assistive technology can select suitable pronunciation. Alternative text must preserve the question without giving away its solution.

Version and scoring integrity

- If translated option text changes meaning, create a new localized version and reassess equivalence. A correction to the common numeric key may affect every language variant; show the impact explicitly.
- Keep original source wording and editorial translation distinguishable in internal provenance. Record translator and reviewer decisions without publishing their private contact information.
- Test long translated text and scripts with different line breaks. A neat English layout does not prove that another language will fit in the same question card.

ACCEPTANCE / PRECAUTIONS

- Switching language retains the current response and maps it to the same option identity.
- A bilingual source creates one item with two reviewed variants.
- A missing or disputed translation prevents an unsupported language promise in paper publication.

Secure untrusted PDFs and AI-assisted extraction

INGEST-17 | FILE AND MODEL SECURITY

Place untrusted documents behind layered controls before any parser or model processes them. Keep academic extraction separate from privileged application actions.

File processing boundary

Validate allowed extension, detected type, signature, size and parser behavior; no single check proves safety. Store sources privately under generated keys, restrict access and use current hardened libraries. OWASP recommends layered upload checks and separate storage or storage outside the webroot [S1]. Treat these as baseline protections, then add the resource limits needed for this workload.

Run PDF parsing, rendering and OCR in isolated workers with minimal permissions, no production database credentials and denied outbound network by default. Bound pages, pixels, decoded image dimensions, CPU time and memory. Reject or quarantine dangerous active features such as embedded scripts, attachments or unexpected external actions according to policy. Keep evidence of rejection without rendering active content in an administrator browser.

Prompt injection and external requests

Treat document text as untrusted data when sending it to an extraction model. A PDF instruction saying reveal the key, fetch this URL or ignore the schema must never become an application instruction. Use a strict output schema, bounded prompts and no privileged tools. Validate all model output before storing draft candidates, and require academic review before publication.

Do not fetch hyperlinks, remote images or document-provided URLs during extraction. If a future URL-import feature is authorized, isolate it behind an allowlisted fetch service with redirect checks, private-address blocking, DNS validation and size limits. Never reuse the application's cloud credentials to retrieve a PDF-supplied location.

Confidentiality and abuse controls

- Redact accidental personal information in sources through a reviewed workflow; keep the original under restricted access. Configure any external OCR or model provider to the agreed retention and confidentiality requirements.
- Use tenant quotas, rate limits and abuse reporting. Do not upload private papers to public scanning services without an explicit data-handling decision.
- Log security events using IDs and sanitized error codes. Never put presigned URLs, document passwords or full private source text in shared operational logs.

ACCEPTANCE / PRECAUTIONS

- [ ] A PDF containing an instruction to expose secrets produces only inert extraction text.
- [ ] A malicious oversized image is stopped by resource limits before affecting the attempt service.
- [ ] A document hyperlink cannot trigger an internal-network request from a worker.

Gate publication, embargoes and content rights

INGEST-18 | RELEASE CONTROL

Enforce publication as a server-side content and rights decision. Make questions, keys, explanations and source downloads independently controlled resources.

Rights and readiness metadata

Require a recorded rights basis before public distribution: owned, licensed, permission_received or another reviewed basis applicable to the content. Keep evidence reference, permitted audience, expiry and restrictions. An uploader assertion can support a private draft, but the platform should not automatically treat every coaching PDF as authorized for hundreds of thousands of users.

Create a readiness report containing manifest integrity, academic approval, verified count, compatible scoring profile, valid assets, rights status and language availability. The publisher sees blockers and documented exceptions. Publication time must use a stored UTC instant with a clear local display; avoid interpreting a timezone-free timestamp differently in browser and worker.

Separate access and release policies

Define question_access, key_access, explanation_access and source_download_access independently. Explanation release may occur after individual submission, after the scheduled test window, on a manual release date or never. A live competitive paper may require delaying explanations even after one student finishes; the policy must specify it explicitly.

Evaluate access on the server for every resource request. Do not ship hidden keys in a JavaScript bundle, encrypted browser field or downloadable source PDF. For scheduled release, avoid caching a permitted solution response across students or time boundaries without including authorization context in the cache policy.

Withdrawal and expiry

- Provide emergency withdraw controls with an affected-paper summary. Decide whether ongoing attempts finish, pause or end under the incident policy; record the decision instead of deleting content mid-attempt.
- When rights expire, prevent new access and stop new assignments according to the licensed terms. Preserve necessary historical records privately subject to the retention policy.
- Add content-report and rights-report workflows. A report opens a tracked case and may quarantine distribution while reviewers inspect evidence; it must not expose the complainant's private information to students.

ACCEPTANCE / PRECAUTIONS

- A student finishing early cannot retrieve explanations before a test-wide embargo ends.
- An expired rights record blocks a new public publication.
- A raw source PDF containing answers is unavailable from student-facing paper routes.

Correct errors with transparent, reproducible rescoring

INGEST-19 | POST-PUBLICATION CORRECTIONS

Build a correction workflow that preserves original evidence while producing trustworthy revised results. Never overwrite completed marks without a visible grading lineage.

Correction classification and impact

Create a CorrectionCase with affected item versions, paper versions, reason, proposed change, academic evidence, severity and reviewer approval. Classify cosmetic, interpretation-affecting, key-affecting and paper-rule errors. A missing minus sign in the stem may invalidate the item rather than merely change the key; the correction reviewer must decide the fair treatment.

Preview affected attempts, completed reports, cohort statistics and assigned future papers. Show whether live attempts encountered the defective version. Freeze the correction decision and resulting policy before batch processing. Do not automatically substitute the latest bank version into every old paper because it seems better.

Rescoring contract

Create grading_run records with original_or_revised, parent_grading_run_id, answer_key_version_id, scoring_config_version_id, correction_case_id and completed_at. Read the original submitted responses; never generate new responses from the revised stem. Use a documented policy for invalidated items, such as excluding their denominator or awarding configured credit, with its effect visible in the result.

Example POST /admin/corrections/case_9/rescore-preview returns {"attempts_affected":1250,"total_changes":117,"max_delta":4,"cohort_recompute_required":true}. Commit creates a durable job using an idempotency key. Chunk work, checkpoint progress and let workers retry without duplicating revised runs. A preview is evidence, not the final committed result.

Student and analytics communication

- Display Original result and Revised result with timestamp, reason and changed questions. Notify affected learners through authorized platform notifications; avoid publishing private student names in a correction notice.
- Recompute affected analytics using revised grading lineage and retain cohort snapshot versions. Percentiles cannot mix corrected and uncorrected marks silently.
- Preserve appeal records and evidence. Allow authorized reviewers to inspect exact historical question assets, key and response when a student disputes a correction.

ACCEPTANCE / PRECAUTIONS

- Repeating a rescoring job creates no duplicate revised grading runs.
- A correction report identifies changed marks and denominator effects clearly.
- Historical original results remain reproducible after the revised result is issued.

Implement search, portable exports and safe migrations

INGEST-20 | CONTENT OPERATIONS

Make the content bank maintainable as it grows. Search and exports must respect visibility, preserve provenance and survive changes to schemas and taxonomy.

Search and deduplication

Provide filters for subject, verified chapter, topic, response type, editorial status, paper membership, source file, reviewer, language and rights status. Search indexed prose and normalized mathematical tokens while keeping structured originals intact. Enforce tenant and role permissions before returning results or counts; an autocomplete suggestion must not reveal a private paper title.

Use exact hashes and similarity candidates to find duplicated content. Distinguish duplicate imports, near-duplicate variants and legitimately reused items. Present source anchors and differences to a reviewer before merging. Retain aliases from merged IDs and preserve historical paper references; deleting an item because two stems look similar can corrupt past attempts.

Export contract and confidentiality

Export a versioned content bundle with manifest.json, items.jsonl, stimuli.jsonl, approved tags, provenance.jsonl and permitted assets. Include schema_version, export timestamp, checksums and source licensing metadata. Separate a question-only export from an administrative archive containing keys and solutions. Require explicit capability for the latter and record the audit event.

Provide CSV for reviewed metadata updates and JSON for full-fidelity content. Escape CSV cells that could be interpreted as spreadsheet formulas. Imports should support dry-run validation, row-level errors, duplicate-ID checks and optimistic concurrency. A portable archive must not include private object credentials or expired presigned URLs.

Migration and retention

- Write migrations that preserve IDs and immutable versions. Test them on representative backups and compare item counts, manifests, keys and source-anchor hashes before and after.
- Keep storage retention rules by artifact type: original source, safe render, OCR draft, published asset and audit record. Remove orphaned derivatives only after confirming they are not referenced by a historical paper.
- Offer a restore drill for one paper and its assets. A database backup alone is insufficient if its manifest points to missing object-storage files.

ACCEPTANCE / PRECAUTIONS

- [ ] An unauthorized search returns no private item counts or title suggestions.
- [ ] Exporting and reimporting a paper reproduces its manifest and grading behavior.
- [ ] A restore drill rebuilds both database references and required content assets.

Prove the pipeline with hostile and academic fixtures

INGEST-21 | ACCEPTANCE AND IMPLEMENTATION PROMPT

Finish this volume with observable end-to-end evidence. Deliver the functioning administrator workflow, not a collection of upload screens backed by fabricated extraction results.

Acceptance corpus and release gates

Create synthetic fixtures for native text, scanned pages, mixed sources, rotated pages, two columns, repeated numbering, shared passages, split-page stems, key tables, missing answers and out-of-order solutions. Add mathematical traps involving a faint minus sign, radical boundary, decimal point, chemical charge and diagram label. Include a bilingual fixture and an asset containing forbidden active content.

For every fixture, record expected items, region mappings, permitted publication state and known failures. Test the actual parser and the administrator correction path. Measure precision and omission rates on the reviewed sample with clear definitions; do not quote a universal 99 percent ingestion accuracy from a handful of clean pages.

Milestones and integration boundaries

Milestone 1: secure upload, private source storage and durable job progress. Milestone 2: page triage, editable segmentation and faithful previews. Milestone 3: typed keys, matched solutions, verified tags and approvals. Milestone 4: immutable paper publication and release controls. Milestone 5: custom practice, correction jobs and portable exports. Each milestone must run against fixtures before the next expands the surface area.

Integrate the exam runner through question-only DTOs in Volume 2, analysis through reviewed taxonomy and grading lineage in Volume 4, and scale/security through Volume 5. Ensure the ingestion system can fail or pause without making exam response saving unavailable. Document any OCR provider dependency and its replacement interface.

Final copy-paste instruction to Emergent

- Implement these requirements with working persistent storage and permission checks. Show a recorded walkthrough importing paired PDFs, correcting one academic discrepancy, publishing a custom paper and retrieving a result.
- Return schema definitions, API examples, worker retry behavior, actual validation outputs and unresolved integration details. Clearly label any placeholder provider and do not portray sample JSON as extracted production content.
- Demonstrate a correction from an original result to a revised result and a blocked attempt to retrieve embargoed explanations. These are completion requirements, not optional future enhancements.

ACCEPTANCE / PRECAUTIONS

- [ ] All 22 page requirements map to implemented behavior, a documented constraint or a named pending dependency.
- [ ] A non-developer reviewer completes the full content workflow using the interface.
- [ ] The acceptance report records actual outputs for clean, ambiguous and hostile fixtures.

Source register and implementation handoff

EVIDENCE AND SCOPE OF THIS SPECIFICATION

Use these references to verify the stated baseline. Requirements and engineering thresholds in this volume are proposed product decisions unless a source explicitly establishes them.

S1 | OWASP File Upload Cheat Sheet

https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html

Checked: 2026-10-11. Primary security guidance; supports layered upload validation, private storage, access control and bounded resource handling. Platform-specific policies in this volume are design requirements.

S2 | PyMuPDF documentation: Text extraction

https://pymupdf.readthedocs.io/en/latest/recipes-text.html

Checked: 2026-10-11. Primary tool documentation; warns about reading order and exposes geometric extraction. Library selection is illustrative and must include a deployment license review.

S3 | PyMuPDF documentation: OCR

https://pymupdf.readthedocs.io/en/latest/recipes-ocr.html

Checked: 2026-10-11. Primary tool documentation; describes OCR as a separate process. It does not establish mathematics recognition accuracy or authorize automatic publication.

S4 | W3C MathML Core

https://www.w3.org/TR/mathml-core/

Checked: 2026-10-11. Primary web standard for mathematical presentation. Accessibility must be tested with actual renderer, browser and assistive-technology combinations.

ACCEPTANCE / PRECAUTIONS

- Before release, retain the approved configuration, source document date, verification owner and test evidence. Re-check any rule that changes for a new exam year.